SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3) Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I Rakkesh karan R(192524253)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ______________________

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing toolusage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks-2

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Assessment Tasks-2

Experiment 1

Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Aim

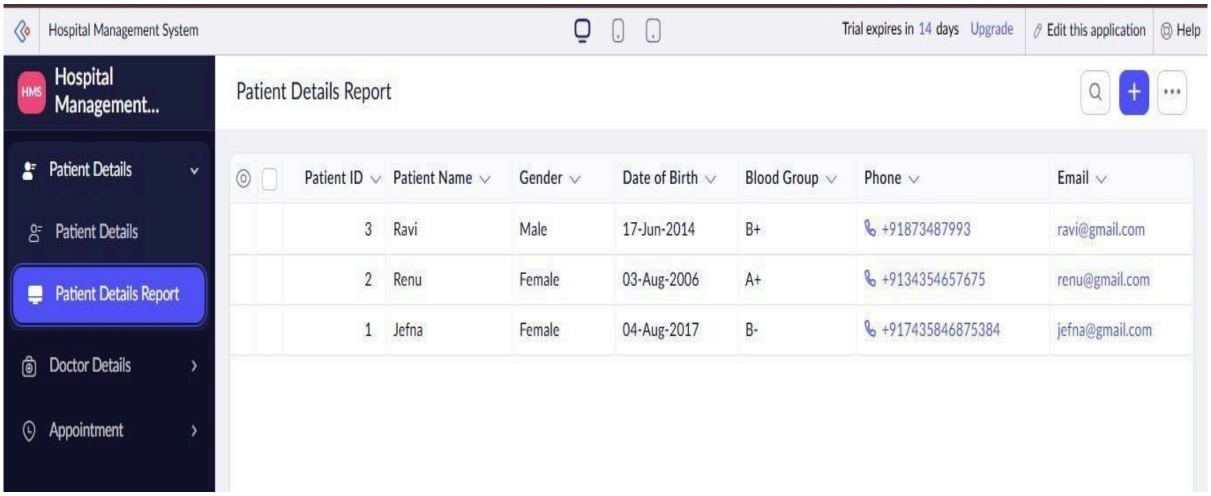
To create a cloud-based **Library Book Reservation System for SIMATS Library** using **Zoho Creator** and demonstrate the **Software as a Service (SaaS)** model.

Procedure

1. Log in to **Zoho Creator** using a valid user account.
2. Click **Create New Application** and name it **Hospital Management System**.
3. Create a **Patient Details** form and add fields such as Patient ID, Patient Name, Age, Gender, Phone Number, Address, Blood Group, and Disease.
4. Create a **Doctor Details** form and add fields such as Doctor ID, Doctor Name, Specialization, Phone Number, Email, and Consultation Fee.
5. Create an **Appointment Details** form and add fields such as Appointment ID, Patient (Lookup), Doctor (Lookup), Appointment Date, Appointment Time, and Appointment Status.
6. Configure the **Lookup** fields to link the Appointment Details form with the Patient Details and Doctor Details forms.
7. Save all the forms and allow Zoho Creator to automatically generate the corresponding reports.
8. Enter sample patient, doctor, and appointment records into the application.
9. Verify that all the entered records are displayed correctly in their respective reports.
10. Access the application through a web browser to demonstrate **Software as a Service (SaaS)** using the cloud platform.

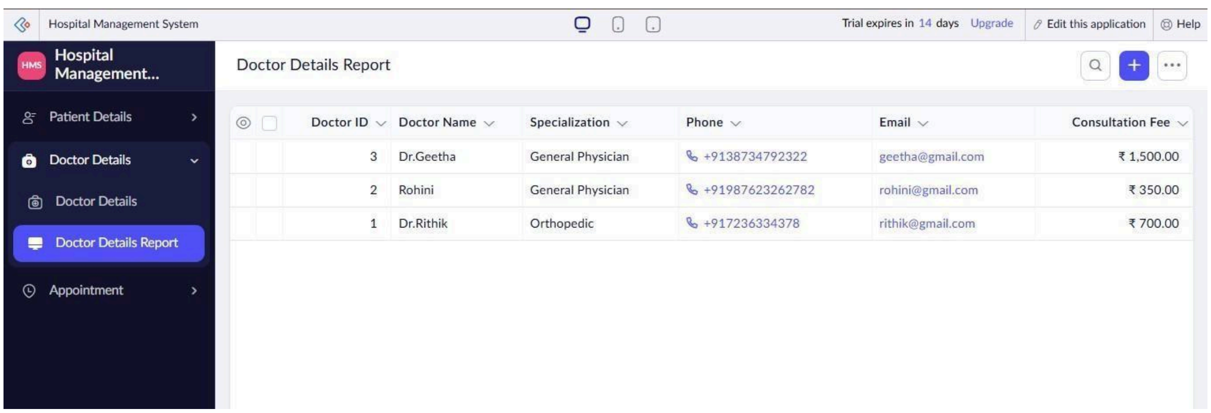
Output

Patient Details:



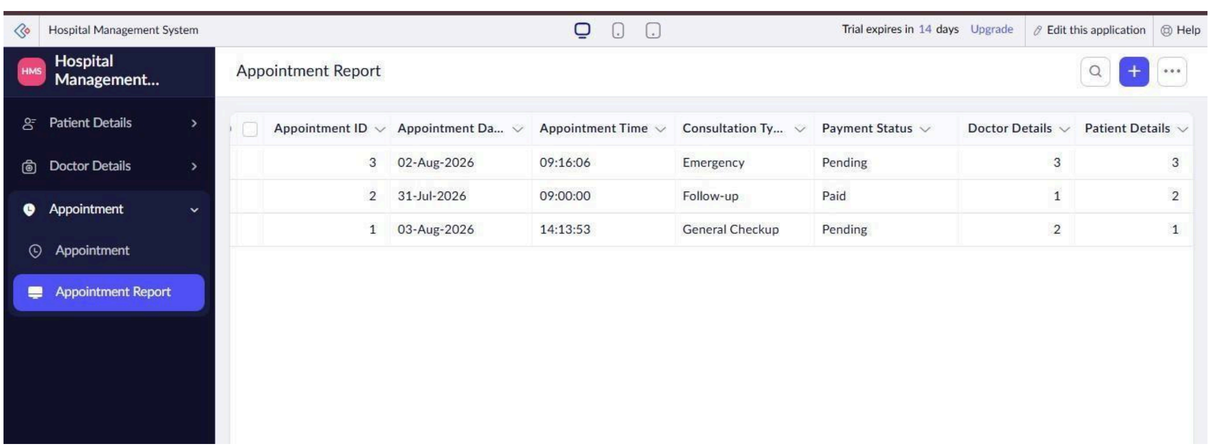
Patient ID	Patient Name	Gender	Date of Birth	Blood Group	Phone	Email
3	Ravi	Male	17-Jun-2014	B+	+91873487993	ravi@gmail.com
2	Renu	Female	03-Aug-2006	A+	+9134354657675	renu@gmail.com
1	Jefna	Female	04-Aug-2017	B-	+917435846875384	jefna@gmail.com

Doctor’s details:



Doctor ID	Doctor Name	Specialization	Phone	Email	Consultation Fee
3	Dr.Geetha	General Physician	+9138734792322	geetha@gmail.com	₹ 1,500.00
2	Rohini	General Physician	+91987623262782	rohini@gmail.com	₹ 350.00
1	Dr.Rithik	Orthopedic	+917236334378	rithik@gmail.com	₹ 700.00

Appointment:



Appointment ID	Appointment Da...	Appointment Time	Consultation Ty...	Payment Status	Doctor Details	Patient Details
3	02-Aug-2026	09:16:06	Emergency	Pending	3	3
2	31-Jul-2026	09:00:00	Follow-up	Paid	1	2
1	03-Aug-2026	14:13:53	General Checkup	Pending	2	1

Result: Thus, a cloud-based **Library Book Reservation System for SIMATS Library** was successfully created using **Zoho Creator**. The application stores book details, student information, and reservation records in the cloud and can be accessed through a web browser, thereby demonstrating the **Software as a Service (SaaS)** model.

Experiment 2

Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Aim

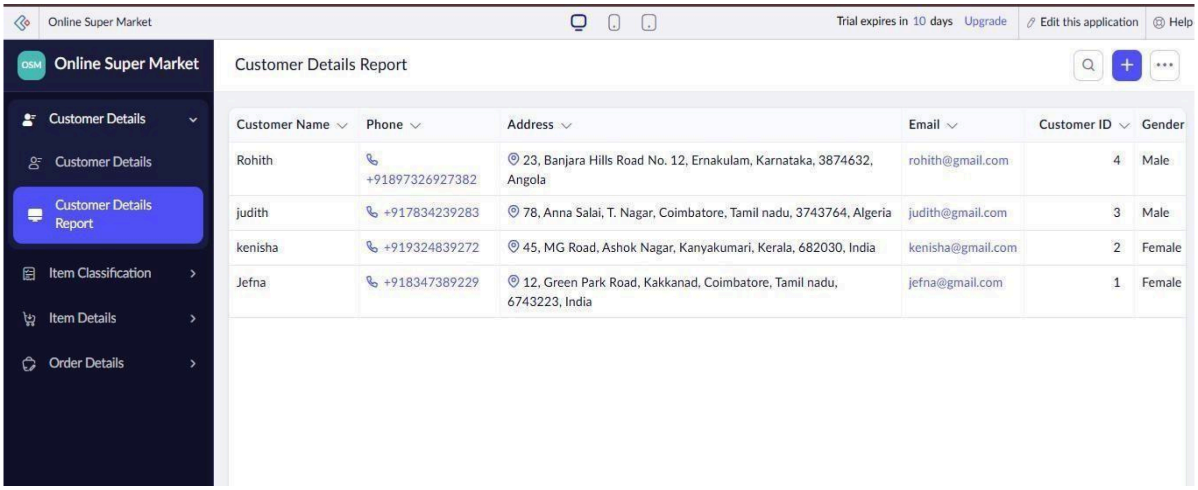
To create a cloud-based **Online Super Market** application using **Zoho Creator** and demonstrate the **Software as a Service (SaaS)** model.

Procedure

1. Log in to **Zoho Creator** using a valid user account.
2. Create a new application and name it **Online Super Market**.
3. Create a **Customer Details** form and add the required fields to store customer information.
4. Create an **Item Classification** form and add the required fields to classify supermarket items.
5. Create an **Item Details** form and add the required fields to maintain item information.
6. Create an **Order Details** form to record customer orders.
7. Configure **Lookup** fields to link the item details with item classification and the order details with customer and item details.
8. Save all the forms and allow Zoho Creator to generate the corresponding reports.
9. Enter sample customer, item, and order records into the application.
10. Verify that all records are stored correctly and displayed in the reports.
11. Access the application through a web browser to demonstrate that it is available as a cloud-based **Software as a Service (SaaS)**.

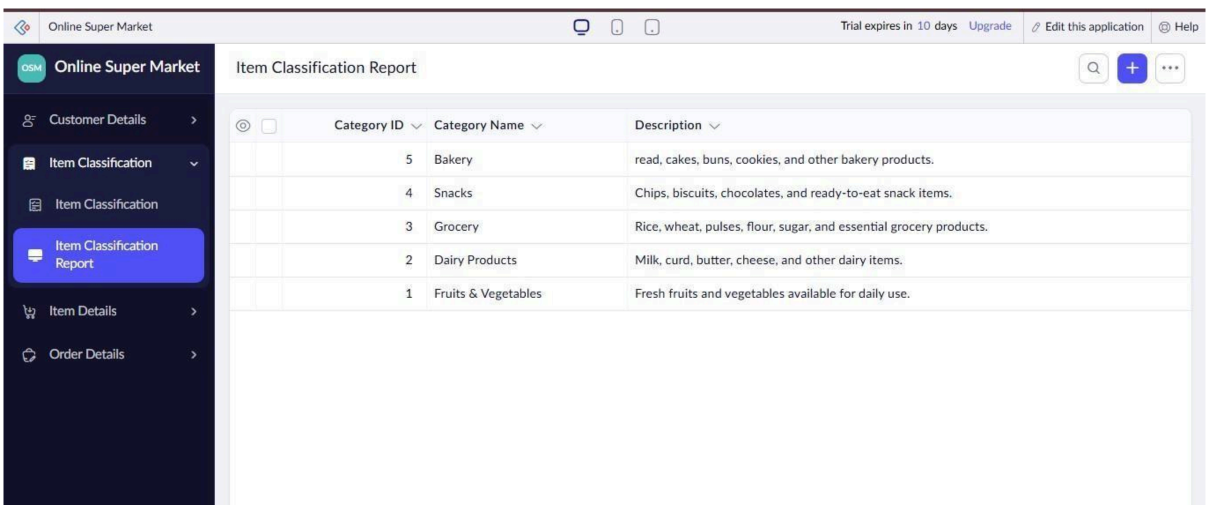
Output

Customer Details:



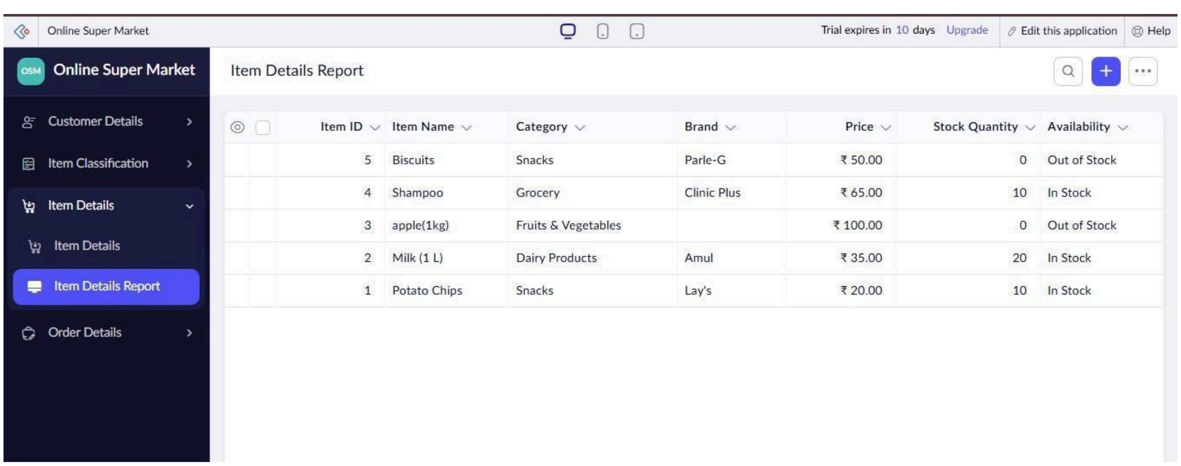
Customer Name	Phone	Address	Email	Customer ID	Gender
Rohith	+91897326927382	23, Banjara Hills Road No. 12, Ernakulam, Karnataka, 3874632, Angola	rohith@gmail.com	4	Male
judith	+917834239283	78, Anna Salai, T. Nagar, Coimbatore, Tamil nadu, 3743764, Algeria	judith@gmail.com	3	Male
kenisha	+919324839272	45, MG Road, Ashok Nagar, Kanyakumari, Kerala, 682030, India	kenisha@gmail.com	2	Female
Jefna	+918347389229	12, Green Park Road, Kakkanad, Coimbatore, Tamil nadu, 6743223, India	jefna@gmail.com	1	Female

Item Classification:



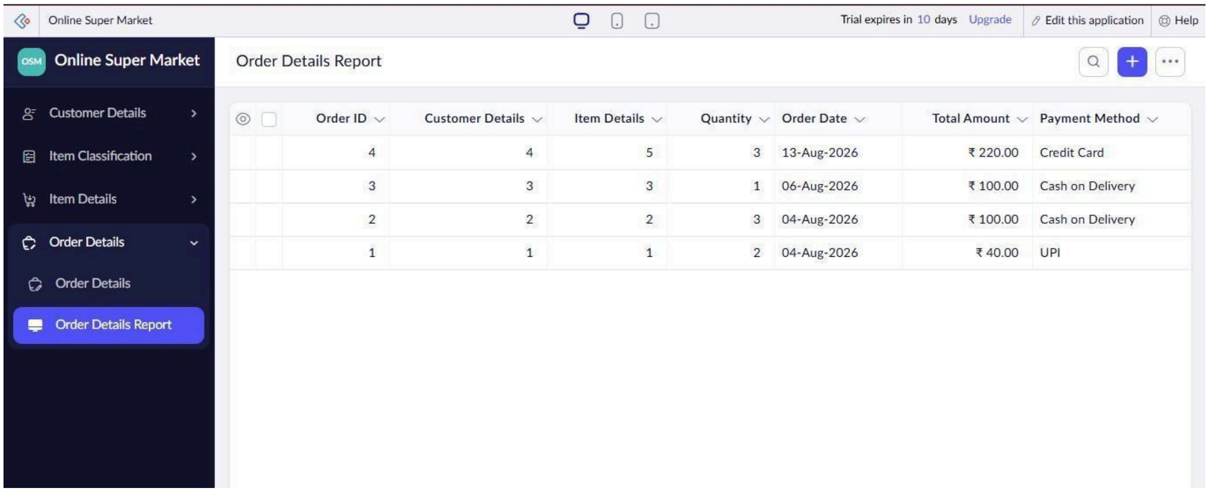
Category ID	Category Name	Description
5	Bakery	read, cakes, buns, cookies, and other bakery products.
4	Snacks	Chips, biscuits, chocolates, and ready-to-eat snack items.
3	Grocery	Rice, wheat, pulses, flour, sugar, and essential grocery products.
2	Dairy Products	Milk, curd, butter, cheese, and other dairy items.
1	Fruits & Vegetables	Fresh fruits and vegetables available for daily use.

Item Details:



Item ID	Item Name	Category	Brand	Price	Stock Quantity	Availability
5	Biscuits	Snacks	Parle-G	₹ 50.00	0	Out of Stock
4	Shampoo	Grocery	Clinic Plus	₹ 65.00	10	In Stock
3	apple(1kg)	Fruits & Vegetables		₹ 100.00	0	Out of Stock
2	Milk (1 L)	Dairy Products	Amul	₹ 35.00	20	In Stock
1	Potato Chips	Snacks	Lay's	₹ 20.00	10	In Stock

Order Details:



Result

Thus, a cloud-based **Online Super Market** application was successfully created using **Zoho Creator**. The application stores customer, item classification, item, and order details in the cloud and can be accessed through a web browser, thereby demonstrating the **Software as a Service (SaaS)** model.

Experiment 3

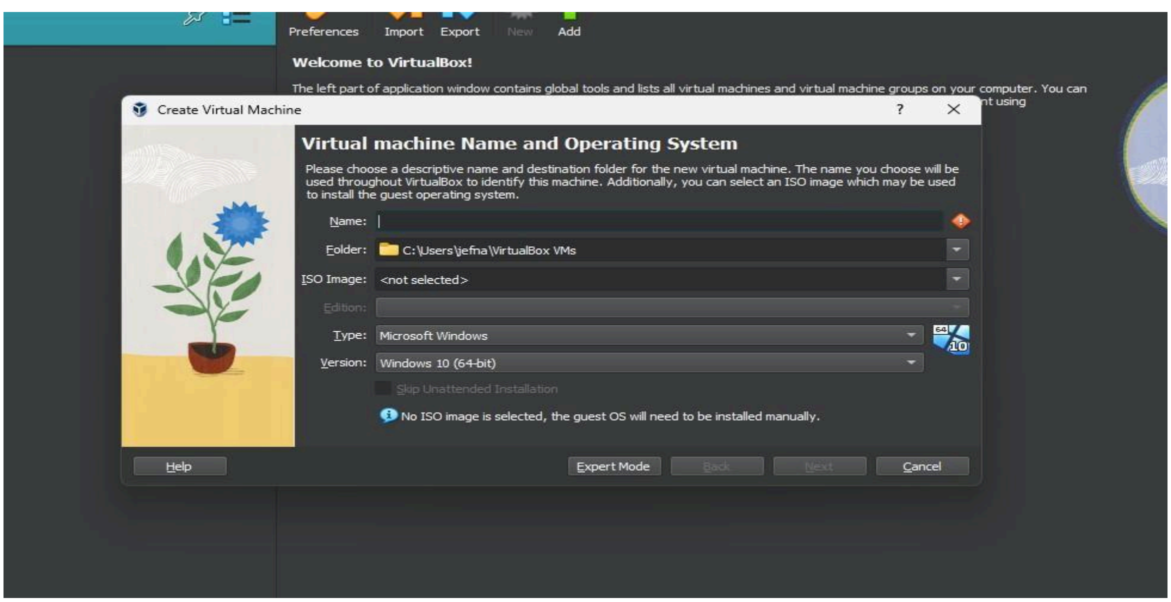
Aim

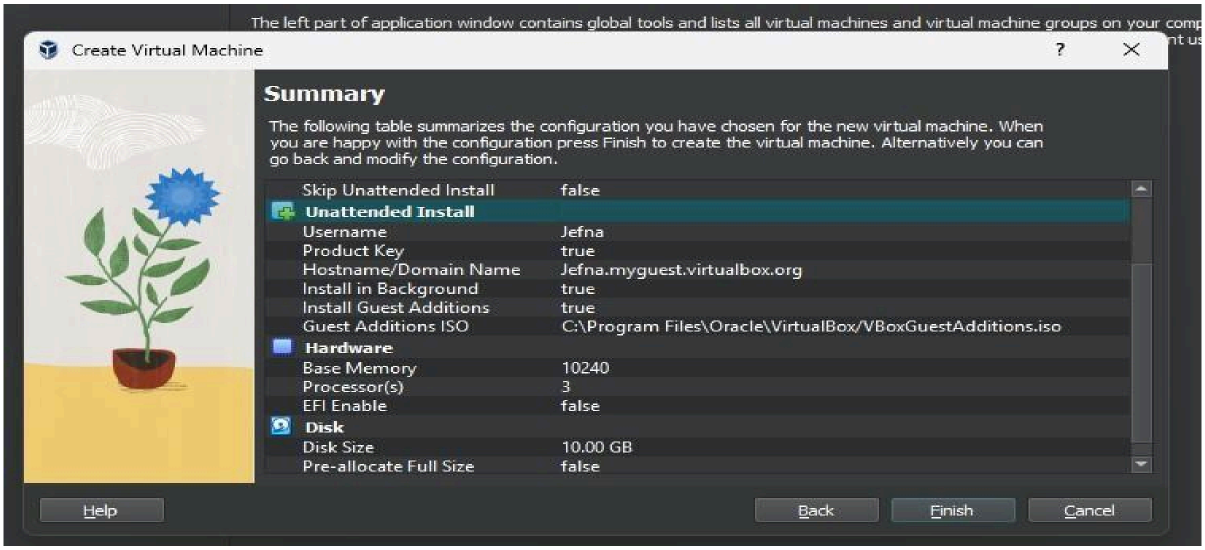
To demonstrate virtualization by installing a Type-2 Hypervisor, creating and configuring a virtual machine with 3 CPU cores, 10 GB RAM, and 10 GB virtual storage using Oracle VirtualBox.

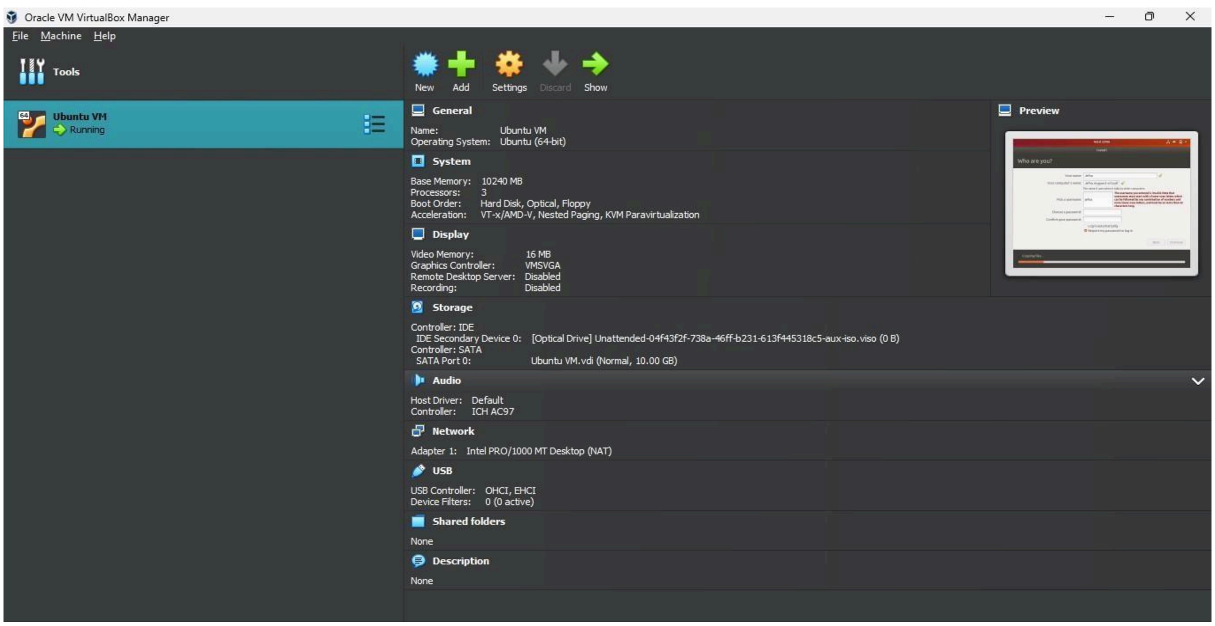
Procedure

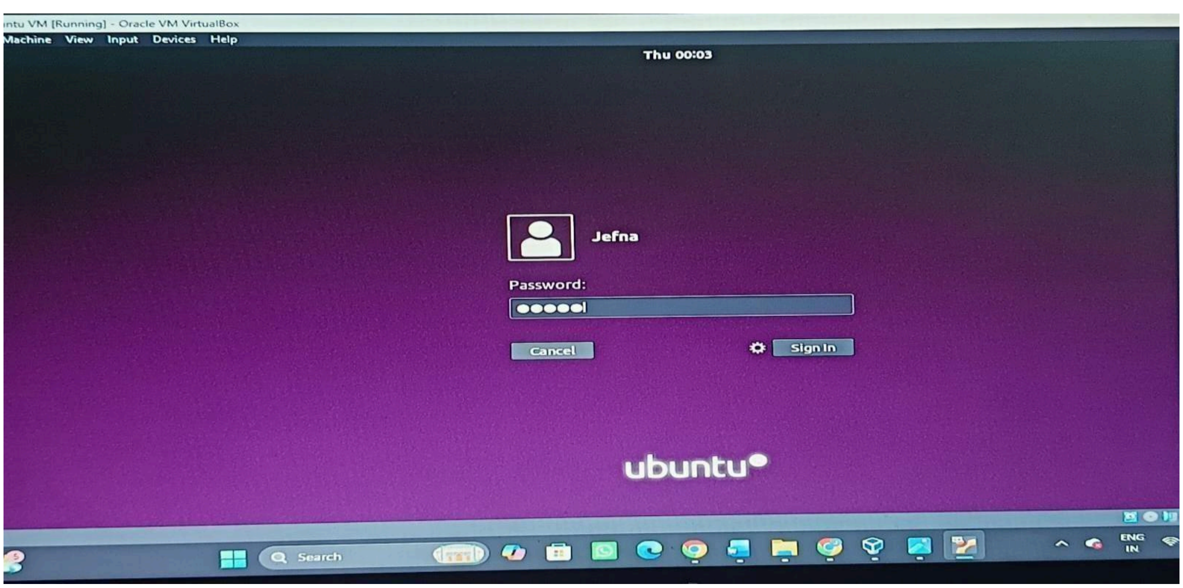
1. Download and install Oracle VirtualBox on the host operating system.
2. Download the Ubuntu Linux ISO image.
3. Open VirtualBox and create a new virtual machine.
4. Select Linux as the operating system.
5. Allocate 10 GB (10240 MB) RAM.
6. Assign 3 CPU cores to the virtual machine.
7. Create a 10 GB virtual hard disk.
8. Attach the Ubuntu ISO file to the virtual machine.
9. Start the virtual machine and install Ubuntu Linux.
10. Verify that the virtual machine boots successfully.

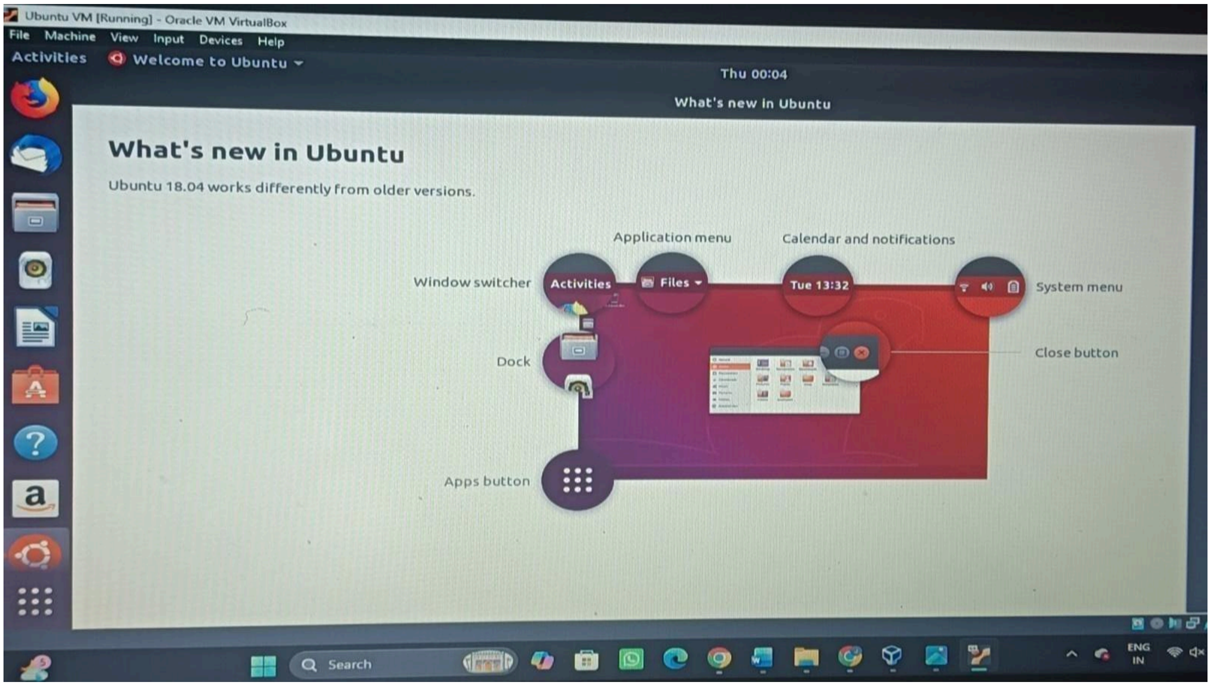
Ouput:

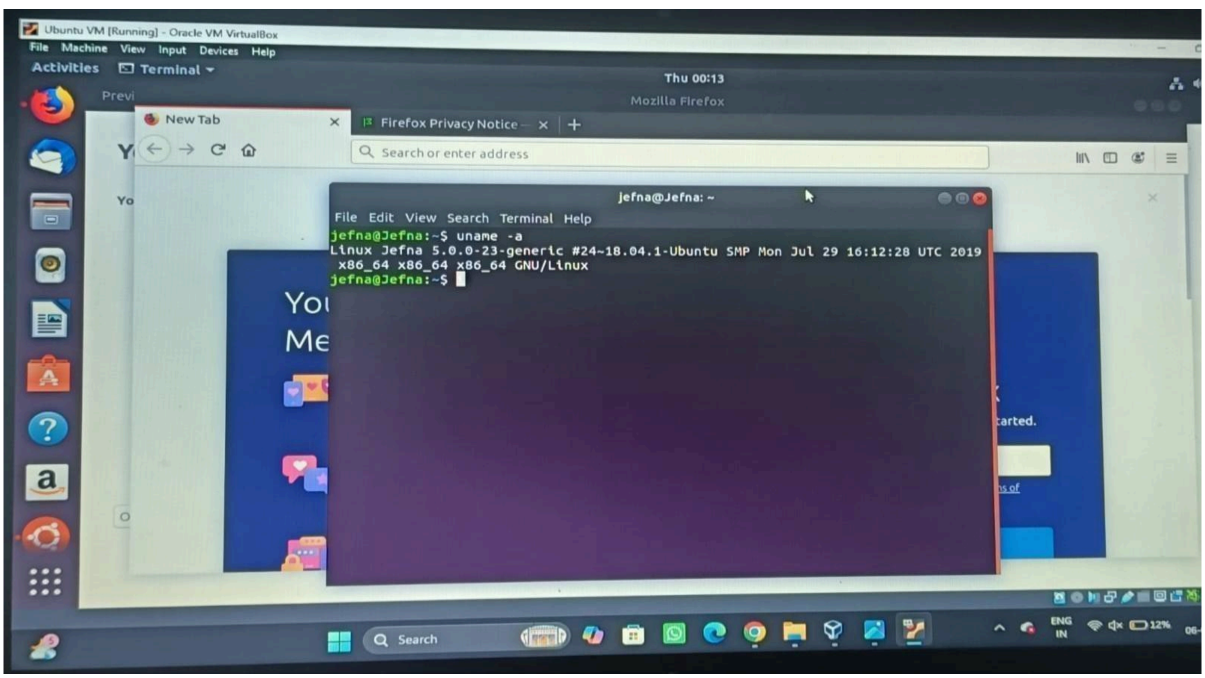



The left part of application window contains global tools and lists all virtual machines and virtual machine groups on your comp
Create Virtual Machine
Summary
The following table summarizes the configuration you have chosen for the new virtual machine. When you are happy with the configuration press Finish to create the virtual machine. Alternatively you can go back and modify the configuration.
Skip Unattended Install false
Unattended Install
Username Jefna
Product Key true
Hostname/Domain Name Jefna.myguest.virtualbox.org
Install in Background true
Install Guest Additions true
Guest Additions ISO C:\Program Files\Oracle\VirtualBox/VBoxGuestAdditions.iso
Hardware
Base Memory 10240
Processor(s) 3
EFI Enable false
Disk
Disk Size 10.00 GB
Pre-allocate Full Size false
Help
Back
Finish
Cancel


Oracle VM VirtualBox Manager
File Machine Help
Tools
Ubuntu VM
Running
New
Add
Settings
Discard
Show
General
Name: Ubuntu VM
Operating System: Ubuntu (64-bit)
System
Base Memory: 10240 MB
Processors: 3
Boot Order: Hard Disk, Optical, Floppy
Acceleration: VT-x/AMD-V, Nested Paging, KVM Paravirtualization
Display
Video Memory: 16 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Secondary Device 0: [Optical Drive] Unattended-04f43f2f-738a-46ff-b231-613f445318c5-aux-iso.viso (0 B)
Controller: SATA
SATA Port 0: Ubuntu VM.vdi (Normal, 10.00 GB)
Audio
Host Driver: Default
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB
USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
None
Preview


ntu VM [Running] - Oracle VM VirtualBox
Machine View Input Devices Help
Thu 00:03
Jefna
Password:
Cancel
Sign In
ubuntu
Search
ENG
IN





Result

Thus, Type-2 virtualization was successfully demonstrated by installing Oracle VirtualBox on the host operating system. A virtual machine was created and configured with 3 CPU cores, 10 GB RAM, and a 10 GB virtual hard disk, and the Ubuntu Linux operating system was installed successfully. Hence, the virtual machine was created and executed successfully.

Experiment 4

Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

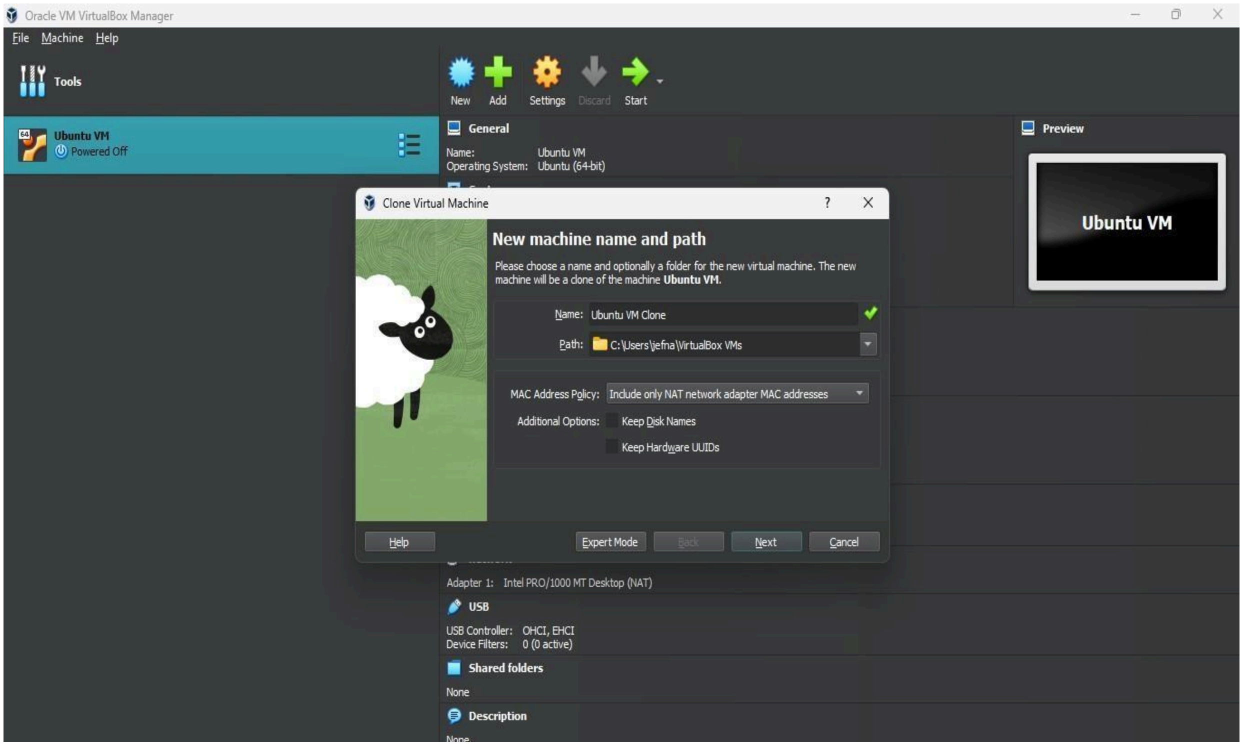
Aim

Clone your existing Ubuntu Virtual Machine and prove that the cloned VM works.

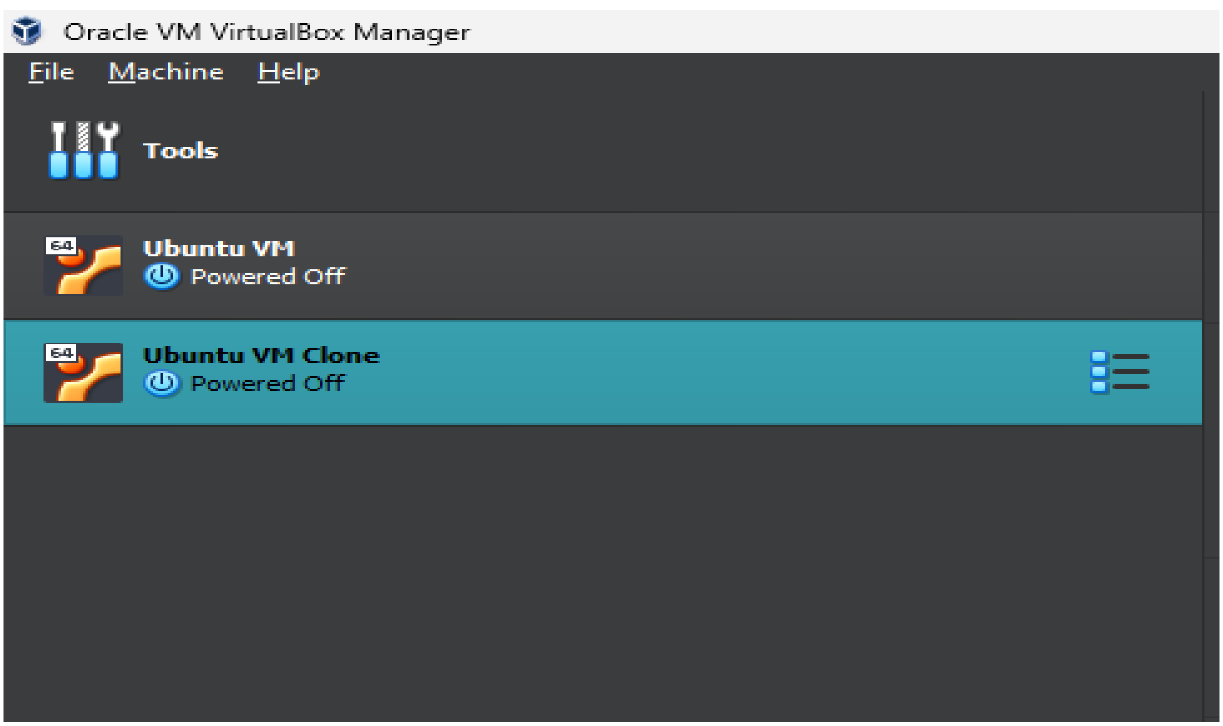
Procedure

1. **Open Oracle VM VirtualBox on the host operating system.**
2. **Ensure the existing virtual machine (Ubuntu VM) is in the Powered Off state.**
3. **Select the Ubuntu VM from the left panel.**
4. **Click Machine → Clone (or right-click the VM and select Clone).**
5. **Enter a name for the cloned virtual machine (e.g., Ubuntu Clone).**
6. **Click Next and choose Full Clone as the cloning type.**
7. **Keep the default MAC Address Policy and click Finish.**
8. **Wait until the cloning process is completed successfully.**
9. **Verify that the cloned virtual machine appears in the VirtualBox Manager.**
10. **Select the Ubuntu Clone and click Start.**
11. **Wait for the cloned VM to boot and verify that the operating system loads successfully.**
12. **Shut down the cloned virtual machine after testing.**

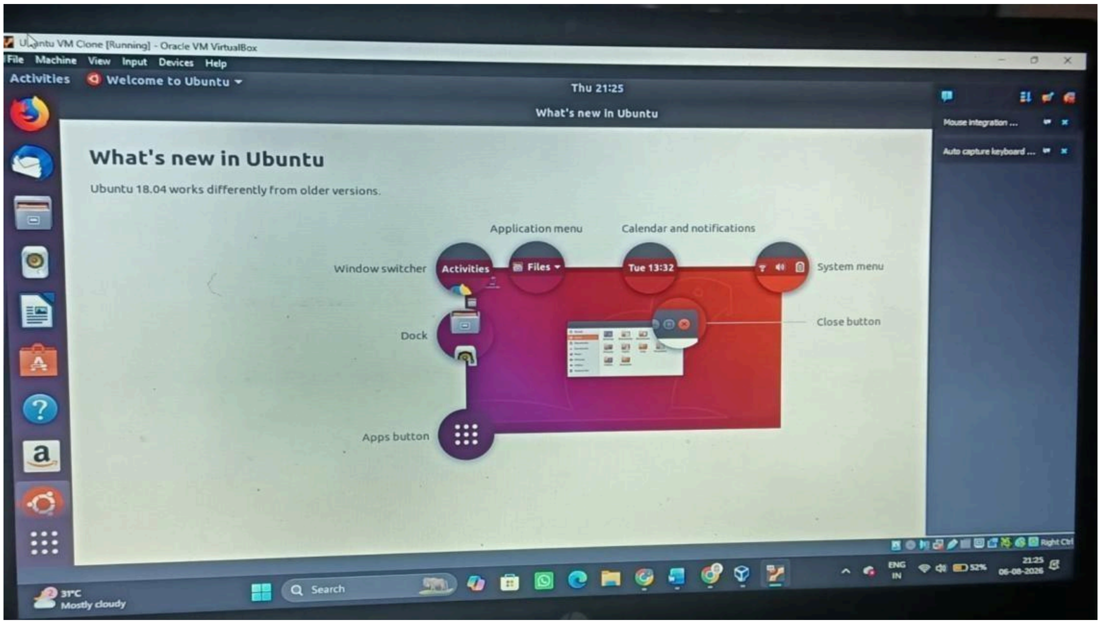
Ouput:

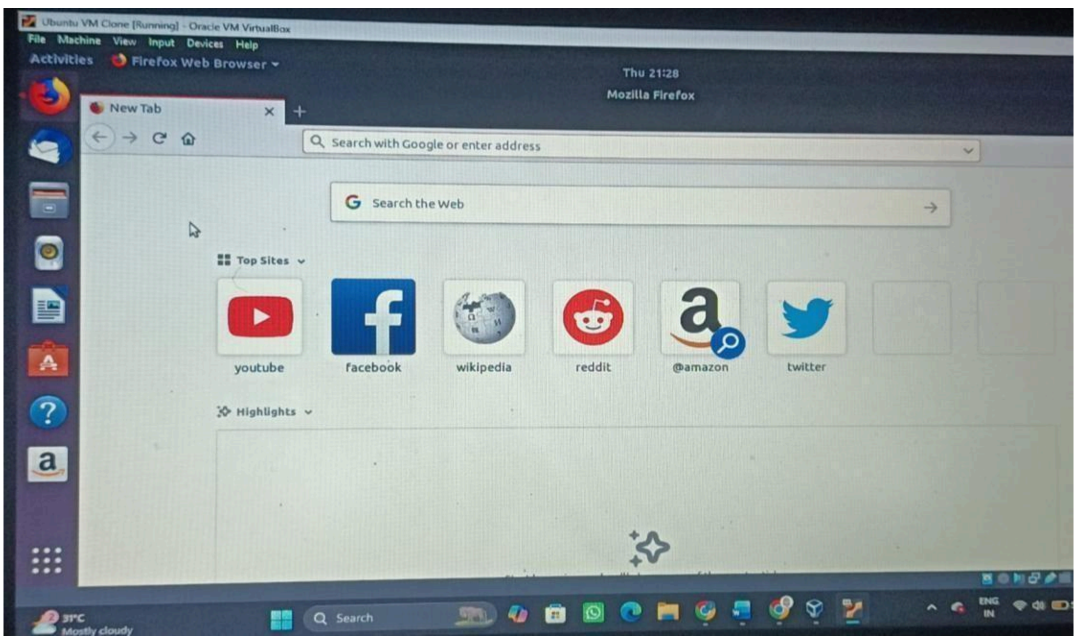


Oracle VM VirtualBox Manager
File Machine Help
Tools
New Add Settings Discard Start
Ubuntu VM
Powered Off
General
Name: Ubuntu VM
Operating System: Ubuntu (64-bit)
Preview
Ubuntu VM
Clone Virtual Machine
New machine name and path
Please choose a name and optionally a folder for the new virtual machine. The new machine will be a clone of the machine Ubuntu VM.
Name: Ubuntu VM Clone
Path: C:\Users\jefna\VirtualBox VMs
MAC Address Policy: Include only NAT network adapter MAC addresses
Additional Options: Keep Disk Names
Keep Hardware UUIDs
Help Expert Mode Back Next Cancel
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB
USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)
Shared folders
None
Description



Oracle VM VirtualBox Manager
File Machine Help
Tools
Ubuntu VM
Powered Off
Ubuntu VM Clone
Powered Off





Result

The virtual machine was successfully cloned using Oracle VM VirtualBox, and the cloned VM was tested successfully by booting into the Ubuntu operating system.